Proceeding of the 2015 IEEE
International Conference on Information and Automation
Lijiang, China, August 2015

Designing a Compact Hexacopter with Gimballed Lidar and Powerful Onboard Linux Computer

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***Abstract*— Building a high-end aerial robot for research is a challenging task. Our aim in this paper is to design a compact hexacopter capable of high payload and has flexible gimbal and powerful onboard computer with $ 500. This hexacopter serves as a flying platform with great customizability, where different sensors can be mounted easily onto the hexacopter for new applications. We discuss our implementation details and preliminary insights during the development process. In addition, we also present our approach on designing a compact gimbal for two cameras, a lidar, and an ultrasonic sensor. Our latest design of gimbal is easy to customize and replicate.**

***Index Terms*— Unmanned aerial vehicles; Unmanned aerial systems; Aerial robotics; Hexacopter design; Gimbal design.**

I. Introduction

Recently, unmanned aerial vehicles (UAV) have been achieving some remarkable performances in the robotic field such as pole acrobatics [1], cooperative flight [2], aggressive maneuver [3], and formation flight [4]. In the near future, we envision UAVs to be our personal assistants, where they can take photos for us, guide us during a tour, and ensure children's safety when they are on the way to school.

Our goal is to design an UAV where it can accompany us and perform various human-robot interaction with us. To realize this goal, we have tried two commercial products in the beginning—Parrot AR.Drone [5] and DJI Phantom [6]. However, we found that both of them cannot handle payload higher than $500\ g$, which is necessary for us to mount additional sensors and onboard computer onto the UAV. A high-end research UAV platform such as AscTec Pelican [7] could handle maximum payload of $650\ g$ but it has a starting price of € 5,195 [8] and is too expensive for our development. To this end, our main motivation is to design an UAV with only $ 500. During our development process, we found many surprisingly well-designed and low-cost UAV components on the market. We are excited to share our development details with other researchers in this paper.

Furthermore, we aim to mount a 2D lidar onto the UAV to perform human detection and obstacle avoidance. Lidar has been *fixed* on the UAV to perform simultaneous localization and mapping (SLAM) and navigation tasks before [9, 10]. However, due to the intrinsic under-actuation characteristic of most conventional UAV, i.e. the 3D position and 3D orientation cannot be controlled independently, the lidar on the UAV cannot scan in its horizontal plane consistently when the UAV is moving. This inconsistency would cause the subsequent algorithms to be more complex. To this end, we also design a gimbal platform for lidar. The gimbal is able to adjust the lidar's pitch and roll orientations so that the lidar can always scan horizontally regardless of the UAV's flying motion. We have designed three compact gimbals with different approaches for the lidar. The latest design is made of 3D printed parts, carbon fiber tubes, and polycarbonate fasteners. It has very high customizability, low weight, and demonstrate superior performance. We would like to share our insights on designing the compact gimbal in this paper.

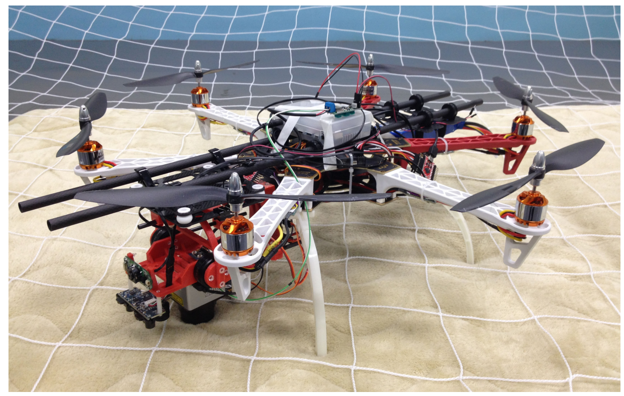

Fig. 1. Our hexacopter has a gimbal at the front side and a battery at the back side. Two cameras, a lidar, and an ultrasonic sensor are mounted on the gimbal. The hexacopter has a diameter of 80 cm and weights 2.5 kg.

Our contribution in this paper is twofold. First, we detail the development process of a compact hexacopter with gimballed lidar and powerful onboard computer. This hexacopter serves as a flying platform with high payload and great customizability, where researchers can mount other sensors like Microsoft Kinect and PS4 stereo camera onto the hexacopter for other applications. Second, we present our insights on designing a compact gimbal for two cameras, a lidar, and an ultrasonic sensor. Our latest design is flexible and easy to replicate. Other researchers can follow our design approach and mount other sensors onto the gimbal easily.

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II. Related Works

A. UAV Hardwares

UAV has many types of configurations, e.g. quadrotor [11], hexacopter [12], octocopter [13], coaxial [14], fixed wing [15], flapping wing [16], and blimp [17]. We chose hexacopter as our platform to achieve fast development since it is one of the most common designs. We also chose hexacopter over quadrotor because it can handle more payload than quadrotor, which is necessary to lift gimbal and lidar.

It is becoming more common to mount 3D depth sensor onto UAV. For instances, researchers have mounted Microsoft Kinect onto their UAVs [18, 19]. Another similar device, Asus Xtion Pro Live, has also been mounted onto UAVs [20, 21]. Both 3D depth sensors are sensitive to sunlight and only work reliably in indoor environment. On the other hand, researchers have mounted Hokuyo UTM-30LX 2D lidar onto their UAVs [9, 18, 22, 23]. A miniature Hokuyo URG-04LX sensor has also been mounted onto UAV [10]. To the best of our knowledge, all 3D depth and 2D lidar sensor are fixed onto the UAVs without gimbal platform.

B. Gimbal Designs

We find that gimbals are not common in the UAV research community. Most gimbals on the market are specially designed for cameras. For examples, DJI company has a plastic gimbal designed for lightweight GoPro camera [24]. Photohigher company has a carbon fiber gimbal designed for large camera up to $1.8\ kg$ [25]. Since both plastic molding and carbon fiber manufacturing are expensive, a creative hobbyist has been using plywood to design his gimbals [26]. We have consulted him for our gimbal design. His gimbals are glued with instant adhesives, lightweight and yet firm. In addition, Do-It-Yourself (DIY) community has also been using 3D printer to design his gimbals [27].

III. System Overview

Designing and developing an UAV is a complex process. The process involves selection of more than 30 components that are dependent on each other. For example, the selection of motor would depend on the weight of hexacopter. Moreover, it affects the selection of propeller and battery, and consequently affect the weight of hexacopter again in a loop. In the followings, we first describe the design of actuator unit in our hexacopter, followed by the processor units, the software backbone, the gimbal design, the sensors and electronics, and the cost summary.[1] As mentioned in the beginning of this paper, our motivation is to design a powerful hexacopter with $ 500. Therefore, we also consider the costs seriously when we select the hexacopter's components.

[1] Due to the limited space, we only list the product names and important characteristics of the components being used in our design.

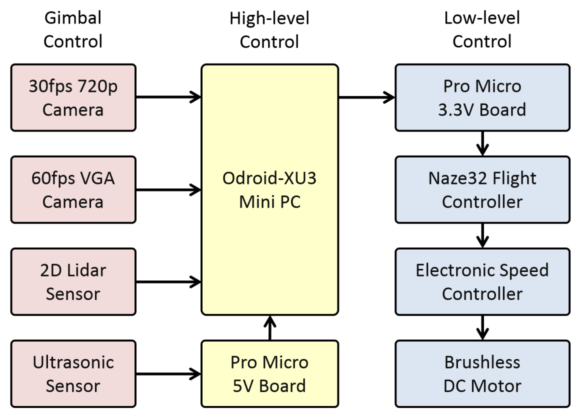

Fig. 2. An overview of the hardware architectures in our hexacopter.

Fig. 2 gives an overview of the hardware architecture in our hexacopter, where the left, middle, and right parts represent sensors on the gimbal platform, the high-level controller, and the low-level controller respectively. Note that the gimbal controller is independent of the hexacopter system and power distribution flow is not included in the figure.

A. Actuator Unit

Motor. We followed online guidance closely during our early design stage. We estimated our hexacopter to have a total weight of $2.4\ kg$, where the body frame, battery, gimbal, and computer & electronics weight $1.5\ kg$, $0.5\ kg$, $0.5\ kg$, and $0.4\ kg$ respectively. Since hexacopter has six identical propulsion units, each motor is expected to produce $400 - 500\ g$ thrust at hovering state. In order to hover the UAV at 50% power, each motor is expected to produce at least $800 - 1000\ g$ thrust at maximum power. We studied many datasheets and selected Turnigy D2830-11 brushless motor for our hexacopter. According to the datasheet, Turnigy D2830-11 brushless motor has maximum thrust of $890\ g$ and consumes $210\ W$. Moreover, it only weights $52\ g$ and is the most lightweight motor that we have found at its price range.

Propeller. The selection of propeller depends on the selected motor and affects the selection of battery. In general, larger and slower turning propellers (hence battery with lower voltage) are more efficient and better for slow flying applications. In contrast, smaller and faster turning propellers (hence battery with higher voltage) are more responsive and better for acrobatic flights. Our main goal is to design a stable hexacopter to be our personal assistant. Therefore, we need no acrobatic flight and selected 1045 propellers, i.e. diameter of $10\ inch$ and pitch of $4.5\ inch$ for our hexacopter. For cost consideration, we use plastic propellers instead of wood or carbon fiber propellers (lighter, stiffer, and more efficient).

Battery. We selected a 3-cell (11.1V) Lithium-Polymer (LiPo) battery for our hexacopter in order to achieve better efficiency. Combining the information of the selected components so far, we expect each Turnigy brushless motor would draw $20A$ at maximum power. This means that the LiPo battery need to have the capacity to supply $6 \times 20A$ continuously. In the hover state, we estimate our hexacopter draws $30A$ in total.[2] At this amount of current, we find that making the hexacopter to fly 30 minutes non-stop at hover state needs a 15,000 mAh LiPo battery, which it weights at least 1.2 kg and is very expensive with the current technology. Therefore, we made some trade-off and design our hexacopter to fly 10 minutes non-stop. Under this design, we selected a 5,000 mAh LiPo battery for our hexacopter. To meet the maximum current supply of $120A$, we selected a Turnigy 3-cell (11.1V) 5,000 mAh 25C (maximum current supply equal to $125A$) LiPo battery.

Electronic speed controller. It is a good practice to select an electronic speed controller (ESC) (used to control brushless motor) that is rated 20% more than the maximum current of the motor. Since the Turnigy D2830-11 motor has maximum current of $20A$, our ESC has to be rate at least $26A$. Furthermore, we chose a new type to ESC in order to have high speed control up to 1 kHz and hence stabilize the UAV better. To this end, we selected Afro ESC $30A$ with SimonK firmware for our hexacopter. For radio controller (RC), one may choose OrangeRX transmitter and receiver since they are low-cost and we only used them in the beginning stage of the development tests.

B. Processor Unit

After the selection of "muscles", we need to select the "small brain" and "big brain" for our hexacopter. We see the low-level flight controller (FC) as the "small brain" in our hexacopter, where it serves as a hub to collect information from all low-level sensors (including accelerometer, gyrometer, magnetometer, barometer, sonar, and GPS), communicates with the RC receiver via serial UART, perform attitude estimation, and send PID control commands to the motor via ESC. On the other hand, we see the onboard powerful computer as the "big brain" in our hexacopter, where it possesses high computation power to perform high-level human-robot interaction in real-time.

Flight controller. There are many high-quality and low-cost FCs on the market such as Naze32, CC3D, Crius, MultiWii, KK2, Naza, Pixhawk, APM 2.6, and etc. In addition to some fundamental requirements such as lightweight, high quality, and low-cost, the FC in our hexacopter has to be powerful enough to perform all the low-level processing reliably. Furthermore, it has to be open-source since we might need to adjust the code in order to include additional sensors in the future. Among the mentioned FCs, Naza is first ruled out since it is not designed for development and is close-source. We then chose Naze32, CC3D, and Pixhawk over Crius, MultiWii, KK2, and APM 2.6 since the former three FCs have state-of-the-art 32-bit processor. While Pixhawk is more powerful, it is much more expensive and heavier than Naze32 and CC3D. To the best of our knowledge, both Naze32 and CC3D are the most lightweight (8 g) and low-cost ($ 25) among the FCs. Eventually, we chose Naze32 over CC3D because Naze32 has conventional pin connector and is easier for our development. Its configuration app in Chrome Store also has minimal design, is easy to use, and yet is powerful enough to meet all our requirements.

Onboard computer. Similar to the FC, there are many high-quality onboard computers on the market. We first tested a Raspberry Pi Model B+ (single core 0.7 GHz) and found that it could not interface with an USB camera (VGA resolution, 30 Hz) smoothly with our basic OpenCV program. The second generation Raspberry Pi (quad-core, 0.9 GHz) might work better but we never test it yet. We also considered LIVA X (dual-core, 1.6 GHz) and Fastwel PC104 (dual-core, 1.7 GHz), and AscTec Mastermind CPU (quad-core, 2.1 GHz). We believe that they are powerful enough for our application but there are too bulky to be mounted onto the UAV. Upon extensive studies and searches, we eventually found a perfect onboard computer for our hexacopter—Odroid-XU3. In fact, it is the most powerful (with A15 quad-core 2.0 GHz and A7 quad-core 1.4 GHz), and yet the smallest onboard computer that we have surveyed (slightly larger than a credit card and only weights 72 g) at a surprisingly low cost of only $ 180. It also has 5 USB ports, 2 UART, and GPIO ports for us to interface with cameras, lidar, Naze32 FC, and other devices. We have since tested Odroid-XU3 intensively. We loaded the board with Linux Ubuntu 14.04 and installed OpenCV and other dependencies in order to test our multi-threading program. The program has four threads—the first thread interfaces with a Logitech camera with VGA resolution at 30 Hz, the second thread interfaces with a PS3 Eye camera with QVGA resolution at 60 Hz, the third thread interfaces with a Hokuyo URG-04LX lidar at 10 Hz, and the last thread interfaces with a Parallax ultrasonic sensor at 50 Hz. Surprisingly, the program can run very smoothly while only utilizing the A15 quad-core. We believe that it can be the "big brain" for our hexacopter for the years to come.

C. Software Backbones

In addition to all the sensor information gathered by the Naze32 FC, we also expect the Odroid-XU3 to be able to interface with multiple devices simultaneously. To this end, we found that multi-threading program is necessary. As mentioned earlier in the Odroid-XU3 section, our software is based on Linux Ubuntu 14.04. We have successfully installed

[2] This is a reasonable estimate since motors generally produce more thrust with little current when operating within half of their maximum capacity.

OpenCV 2.4.9 and Hokuyo library 1.1.8 onto Odroid-XU3. We have been writing our programs in C++ and have been using POSIX threads programming. We find that it is also possible to install OpenGL (for graphics processing) and OpenNI (to interface with Microsoft Kinect or Asus Xtion Live) onto Odroid-XU3. In order to test the hexacopter control during the development, we also developed a C program to receive commands from Wiimote via Bluetooth and send commands to Naze32 via a serial interface. We tested the program and it works smoothly at 250 Hz without failure for more than 30 minutes.

D. Gimbal Design

Gimbal structure. In the beginning of the gimbal design, we purchased an off-the-shelf gimbal designed for GoPro camera. After making some modifications, we mounted a downward camera and an ultrasonic sensor onto the gimbal (Fig. 3 (a)). We found that it is very hard to maintain the balance of the gimbal with this approach. Without a good balance, gimbal would not work effectively and smoothly. Then, we proceeded to design a lightweight gimbal from scratch with balsa wood (Fig. 3 (b)). It worked as expected but the building process was very time-consuming. Furthermore, we speculate that it would not be stiff enough to hold heavier sensor such as lidar (160 g in our case).

In our third design, we used a 3D printing method and designed our gimbal with Tinkercad [28]. With the intuitive interface of Tinkercad, we designed our customized gimbal to hold various sensors, motors, and controller precisely. We printed our gimbal structures and assembled the parts with polycarbonate fasteners to reduce gimbal weights. Similar to the previous designs, the most important process is to balance the gimbal. We have adjusted the center of gravity and position of gimbal motors for a few times. The resulting gimbal is impressive (Fig. 3 (c)). It is easy to mount all the sensors, motors, and controller. It is also compact, lightweight, and stiffer than we have thought. More importantly, the design process is very flexible for customization. With a basic design, a new gimbal can be produced within one day, which is hardly possible with the first two approaches.

Gimbal controller. Instead of designing a gimbal controller from scratch, we opted to use off-the-shelf controller to save development time. Upon extensive survey, we found that AlexMos (8-bit version) and Martinez gimbal controllers are our best options, in term of functions, cost, and reliability. We have tested both controllers. Both controllers are working after similar tuning process but we find that AlexMos is easier to use. In particular, we had no problems to configure AlexMos parameters via the USB connection every time. AlexMos controller also produces less high frequency noises upon fine tuning and has reverse voltage protection. We eventually chose AlexMos controller for our hexacopter, even though it is 2 times more expensive than Martinez controller.

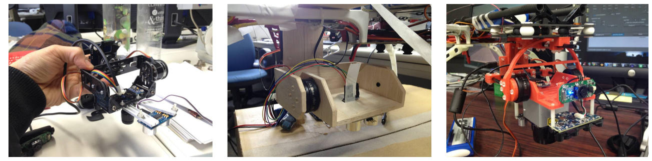

Fig. 3. We have designed three gimbals, where each of them is made of (a) carbon fiber plates, (b) balsa wood, and (c) ABS plastic respectively. We chose 3D printing method in our latest design as the gimbal prototype is compact, stiff, lightweight, and very flexible for customization.

E. Sensors and Electronics

Sensors. In addition to the sensors that are integrated with Naze32 FC and AlexMos gimbal controllers, we added four sensors onto the gimbal platform—a frontal Logitech camera (1080p, 30 Hz), a downward PS3 Eye camera (VGA, 60 Hz), a Hokuyo URG-04LX 2D lidar (682 measurements in $240°$ range, 10 Hz), and a Parallax PING))) ultrasonic sensor (3 meters, 50 Hz). All four sensors are mounted on the gimbal platform in order to decouple the sensor measurements from the UAV's flight motion. With the gimbal, we can use the sensor data for high-level processing directly.

We aim to use the downward PS3 Eye camera to compute optical flow with Odroid-XU3 in our future work in order to perform accurate position holding and navigation. Compared to the popular ADNS3080 mouse-based optical flow sensor, PS3 Eye camera has higher image resolution, larger field of view, and is less expensive. In addition, we also chose the Parallax ultrasonic sensor over the HC-SR04 sensor. We tested both sensors and found that Parallax ultrasonic sensor can achieve measurement rates of 40 Hz (an object that is 2 m away) and works more reliably.

Electronics. In order to reduce the hexacopter's weight, we draw power from the main battery to Odroid-XU3 (5V) and gimbal (9V) via off-the-shelf switching regulators. We have tested the switching regulators mentioned in Table I and made sure that both the Odroid-XU3 and gimbal (including controller and motors) works reliably in more than 2 hours.

F. Cost Summary

We summarizes the development cost of our hexacopter in Table I. The costs of the additional frontal Logitech camera and Hokuyo lidar are not included in our calculation. Our current first prototype costs $ 992, which is almost two times of our goal ($ 500). However, we believe our second prototype could be much cheaper because we could potentially reduce some components such as RC transmitter, RC receiver, and Pro Micro boards in our second design. It is also worth noting that the components marked with $*$ in Table I have cheaper counterparts on the market. In our second design, we believe we could further reduce the development cost to less than $ 750.

TABLE I

COST SUMMARY OF HEXACOPTER IN A NUTSHELL. OUR CURRENT FIRST PROTOTYPE COSTS $ 992. THE COMPONENTS MARKED WITH * HAVE COUNTERPARTS ON THE MARKET AND COULD DRIVE DOWN THE DEVELOPMENT COST TO LESS THAN $ 750 AT THE TIME OF WRITING.

Hexacopter Basics	US$	Onboard Computer	US$	Electronics	US$	Gimbal	US$
DJI hexacopter flame wheel	60	*Odroid-XU3 board	179	Naze32 Acro controller	25	*AlexMos controller & IMU	106
Helipal camera positioning kit	55	*eMMC module	39	9V voltage regulator	25	Gimbal motors	55
Turnigy 5000 mAh battery	25	Power cable	2	5V voltage regulator	25	5 mm carbon fiber tube	10
Turnigy motors & propellers	84	Bluetooth dongle	7	*Pro Micro board 5V	20	*2 mm carbon fiber plate	30
Afro electronic speed controllers	66	Wiimote controller	34	*Pro Micro board 3.3V	20	Damping rubber & bearing	10
*OrangeRX transmitter & receiver	67	PS3 Eye camera	8	*Ultrasonic sensor	30	Polycarbonate fasteners	10
	357		269		145		221

IV. EXPERIMENT RESULTS

A. Flight Time Test with Spektrum Controller

Without the gimbal platform and onboard computer, the hexacopter has 1.8 kg. We first tested the strength of the body frame by putting dummy weights onto the hexacopter. We made sure the hexacopter is well balanced and used Naze32 default PID parameters without fine tuning. We performed tests in a small indoor room and due to the disturbance caused by the wind generated by the hexacopter itself, we need to maintain its position manually during the test. In one test, the hexacopter has 2.2 kg and is able to hover steadily at 50% motor power. In another test, we added dummy weights until the hexacopter has 2.5 kg and the hexacopter is able to hover steadily at 54% motor power.

In order to test the flight time of hexacopter, we tightened the hexacopter on the top of a heavy plate. We have fully charged the 5000 mAh LiPo battery before the tests. The battery power is monitored by using a typical battery checker. We then powered the hexacopter at 54% motor power and measured a flight time of up to 9 minutes when the battery remains 5% power. In the second test, we powered the hexacopter at 58% motor power and measured a flight time of up to 8 minutes when the battery remains 5% power. Fig. 4 shows the two flight time plots with 5 degrees polynomial fitting. Both plots have nonlinear increases of flight times with decreasing battery capacity, where the battery capacity drops faster in beginning of the flights.

B. Flight Test with Wiimote Controller

After making sure that the hexacopter frame works properly, we installed Odroid-XU3 at the top of the hexacopter. We then run a program on Odroid-XU3 to communicate with Wiimote via Bluetooth and send serial commands to Naze32 via Pro Micro 3.3V. This program test had a few important purposes. First, we made sure that the Odroid-XU3 could send serial commands to Naze32 reliably, which is essential for high-level control. Moreover, while the conventional RC transmitter and receiver communicate at 50 Hz, we found that Odroid-XU3 and Naze32 is able to communicate reliably at 250 Hz. With this control rate, Odroid-XU3 can perform high speed control in our future applications. Second, for safety reasons, Wiimote acts as an emergency controller at the later stage of our autonomous flight tests.

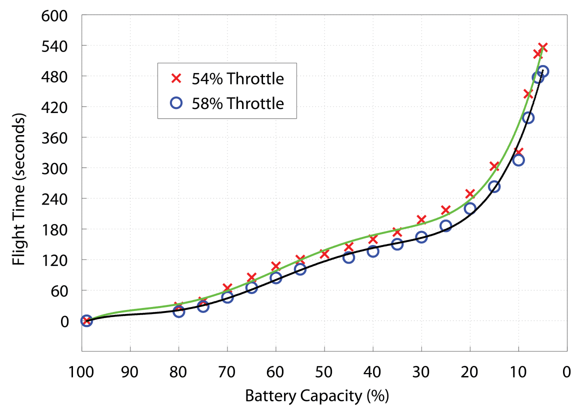

Fig. 4. Flight times of our hexacopter at 54% and 58% motor powers at different battery capacity levels with 5 degrees polynomial fitting.

V. CONCLUSION

Summary. In this work, we designed a compact hexacopter capable of high payload and has flexible gimbal and powerful onboard computer with less than $ 1,000. We also discussed our implementation details extensively. We have tested our prototype and made sure that it can handle additional payload up to 700 g and able to hover steadily up to 8 minutes. The hexacopter is also equipped with a powerful onboard Linux computer. Currently, the onboard computer is able to run a C++ multi-threading program to interface with two cameras, one lidar, and one ultrasonic sensor smoothly. In addition, we also present our approach and insights on designing a compact gimbal that is easy to customize and replicate.

Future works. We aim to implement some basic control routines such as height-maintaining flight, position-holding hover, and accurate navigation control for our hexacopter. Together with the designed gimbal platform and powerful onboard computer, we are looking forward to realize some high-level functions such as human detection with hybrid camera and lidar data, obstacle avoidance, and SLAM applications.

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